

Experimental Trajectory Tracking of a Differential-Drive Wheeled Mobile Robot Using Backstepping Control with Sliding-Mode-Inspired Compensation via ESP-NOW

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Abstract—This paper presents the implementation and experimental evaluation of a velocity-level Backstepping trajectory-tracking controller augmented with bounded sliding-mode-inspired nonlinear compensation for a differential-drive wheeled mobile robot. The low-cost platform combines an STM32F411 wheel-speed controller, ESP32/ESP-NOW communication, ROS 2 high-level control, and an overhead AprilTag camera fused with wheel odometry through an Extended Kalman Filter. Backstepping, conventional SMC, and the compensated controller are evaluated on circular, figure-eight, and square references. The currently selected physical runs show comparable Backstepping and BSMC performance on the circle and square, while the selected pure-SMC square pilot exhibits larger corner transients. Manual circle-displacement pilots further expose the importance of global relocalization and inner wheel-speed behavior. Because the figure-eight, square-SMC, and disturbance datasets are selected pilots rather than balanced locked-gain repeats, they are reported as preliminary physical evidence and not as final statistical validation.

Index Terms—Backstepping control, sliding-mode-inspired compensation, wheeled mobile robot, ESP-NOW, AprilTag localization, Extended Kalman Filter.

I. INTRODUCTION

Trajectory tracking of differential-drive robots is affected not only by the nonholonomic kinematics but also by actuator deadzones, wheel asymmetry, wireless timing variation, encoder quantization, and localization drift. Backstepping provides a constructive nominal tracking law, whereas bounded nonlinear compensation can increase robustness without discontinuous switching. This work evaluates that combination on a low-cost physical platform and compares it with independently configured Backstepping and SMC baselines.

II. CONTROLLER AND PLATFORM SUMMARY

For pose $q = [x, y, \theta]^\top$, the kinematic model is

$$\dot{x} = v \cos \theta, \quad \dot{y} = v \sin \theta, \quad \dot{\theta} = \omega. \quad (1)$$

Using body-frame errors e_x , e_y , and e_θ , the compensated commands are

$$\begin{aligned} v_{\text{cmd}} &= v_d \cos e_\theta + k_1 e_x + K_{s1} \text{sat}(e_x/\varphi_1), \\ \omega_{\text{cmd}} &= \omega_d + v_d(k_2 e_y + k_3 \sin e_\theta) + K_{s2} \text{sat}((e_\theta + c e_y)/\varphi_2). \end{aligned} \quad (2)$$

$$(3)$$

The controller runs on a ROS 2 host. Velocity commands and wheel telemetry are exchanged through an ESP-NOW bridge, while the STM32 closes the two wheel-speed loops. An overhead camera observes a 0.15-m AprilTag and supplies absolute pose corrections to a custom EKF.

III. EXPERIMENTAL METHOD

The physical tests use circle, figure-eight, and 2-m square references. Camera-derived tracking errors are computed after rigid local-frame alignment, without temporal smoothing. Nominal experiments compare Backstepping, BSMC, and SMC. Manual disturbance pilots are restricted to the circle and use two displacement orientations. A run is considered recovered only when its position error returns below the fixed threshold and remains there for the prescribed hold interval.

IV. PHYSICAL EXPERIMENTAL RESULTS

A. Experimental Data Scope

The physical experiments were performed on the STM32F411–ESP-NOW–ROS 2 platform using overhead AprilTag localization fused with wheel odometry. Figure 1 shows the camera view and the marked robot in the test area. All reported errors were recomputed from the camera pose after rigid alignment to the local trajectory frame; no temporal smoothing was applied during post-processing.

The currently available data do not form a balanced final validation set. The circular nominal runs contain five laps per controller; the selected square runs contain three BSMC laps, two Backstepping laps, and one SMC tuning lap; and the figure-eight comparison uses the best available pilot run for each controller. Consequently, the results below are presented

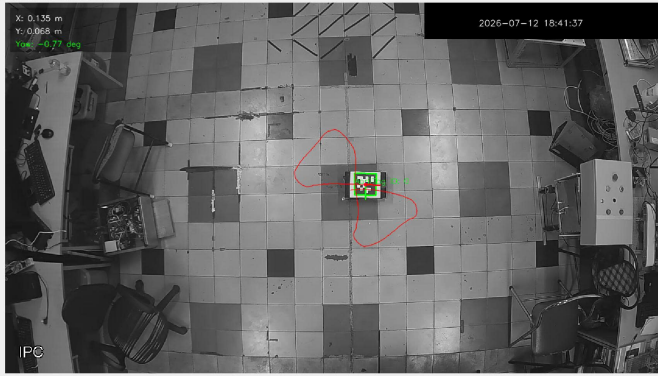


Fig. 1. Overhead-camera view of the physical test area. The AprilTag attached to the robot provides the absolute pose measurement used by the EKF.

TABLE I
SELECTED PHYSICAL TRACKING RUNS

Path	Controller	Position RMSE (cm)	Heading RMSE (deg)
Circle	Backstepping	2.12	1.28
	BSMC	2.27	1.52
	SMC	2.53	2.00
Eight	Backstepping	3.16	4.69
	BSMC	2.68	3.33
	SMC	2.66	5.32
Square	Backstepping	3.22	22.26
	BSMC	3.04	22.59
	SMC [†]	8.95	24.12

[†]Tuning pilot; not a locked validation run.

as an experimental pilot comparison. In particular, the SMC square run is tuning data and is not relabeled as validation data.

B. Nominal Trajectory Tracking

Figure 2 compares the measured trajectories for the three reference families. On the circle, all controllers remain close to the reference, with nominal Backstepping giving the lowest selected position RMSE. On the figure-eight, BSMC and SMC give similar position RMSE, whereas BSMC gives the lowest heading RMSE. For the sharp-corner square, BSMC and Backstepping are close, while the selected pure-SMC pilot exhibits larger outward corner transients.

The large overall square heading RMSE values arise primarily from the instantaneous 90° reference-heading changes at the vertices. To separate corner transients from established straight-edge behavior, Table II reports metrics only on the detected straight portions. BSMC gives the lowest straight lateral RMSE, while Backstepping gives the lowest straight heading RMSE. The SMC pilot requires substantially more angular correction and does not return to the new edge as quickly after each corner.

C. Manual Disturbance and Relocalization Pilots

The manual disturbance protocol was applied only to the circular trajectory. The robot was paused, physically displaced, and then released while the controller continued from the same

TABLE II
STRAIGHT-EDGE METRICS FOR THE SELECTED SQUARE RUNS

Controller	Lateral RMSE (cm)	Lateral bias (cm)	Maximum $ e_y $ (cm)	Heading RMSE (deg)
Backstepping	3.44	3.28	5.35	2.11
BSMC	3.26	3.07	5.53	2.24
SMC [†]	8.94	6.79	19.19	7.90

TABLE III
SELECTED CIRCLE DISTURBANCE PILOTS

Yaw	Controller	Peak increment (cm)	IAE ₁₀ (ms)	T _{rec} (s)
0°	Backstepping	30.17	1.339	6.88
	BSMC	28.37	1.279	7.24
	SMC	19.24	1.366	NR
90°	Backstepping	52.66	3.451	NR
	BSMC	54.19	3.778	NR
	SMC [‡]	74.40	3.516	NR

NR: recovery criterion not reached. [‡]Safety-censored run.

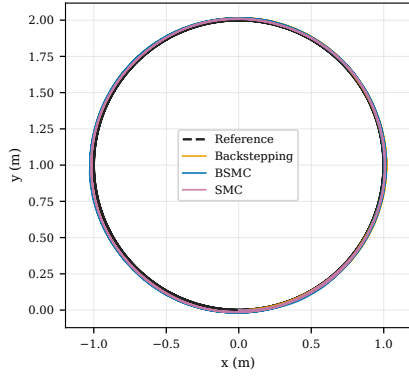
reference phase. Two displacement orientations were recorded: nominal yaw (0°) and an approximately 90° yaw change. These trials were designed as relocalization and safety pilots, not as balanced repeat statistics.

The yaw-0° pilot shows successful recovery for Backstepping and BSMC, whereas the selected SMC run does not meet the recovery criterion. None of the selected yaw-90° trials meets the same criterion. The result demonstrates that large pose relocalization is a substantially harder closed-loop condition than nominal tracking. It does not establish a statistical ordering because only selected pilot runs are available and the applied manual displacements are not identical.

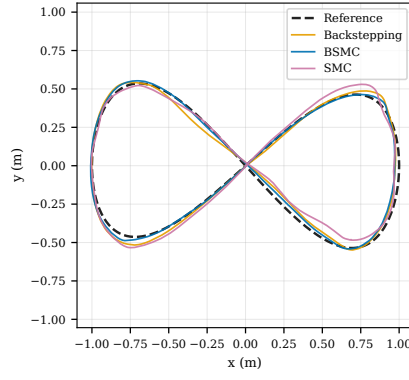
D. Implementation Observations and Validity

The experiments revealed two implementation effects that are not visible in the ideal kinematic simulation. First, absolute camera measurements must be allowed to re-enter the EKF after a sustained large innovation; otherwise the filter can continue from an encoder-only phantom pose after the robot is physically displaced. Second, square-corner performance depends strongly on the inner wheel-speed loop. For the same nominal corner command, a stalled inner wheel increases the effective turning radius and leaves less straight distance for lateral-error recovery. These effects motivate reporting camera-derived errors and actuator telemetry in addition to controller commands.

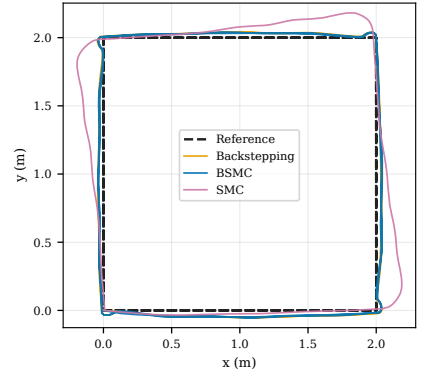
The nominal circle data are the most balanced set presently available. The figure-eight and square comparisons remain selected pilots, and the disturbance data are manual safety trials. Final publication claims requiring mean and standard deviation must therefore be based on a future locked-gain, equal-repeat validation campaign rather than on the selected-run values in this section.



(a) Circle: five-lap selected runs.

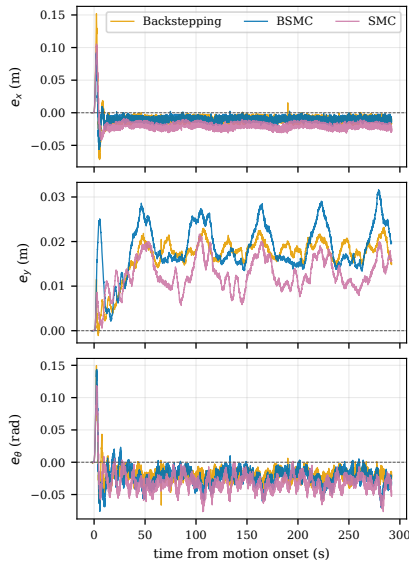


(b) Figure-eight: selected pilot runs.

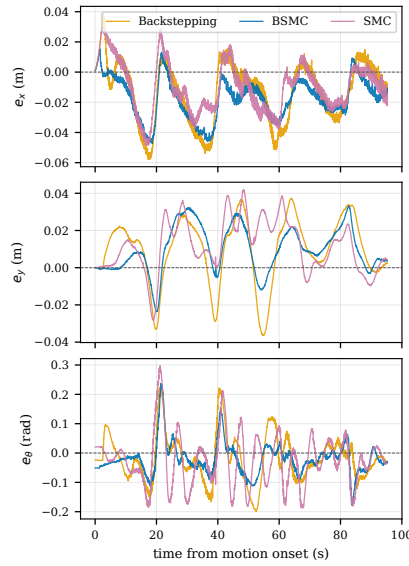


(c) Square: selected runs with unequal lap counts.

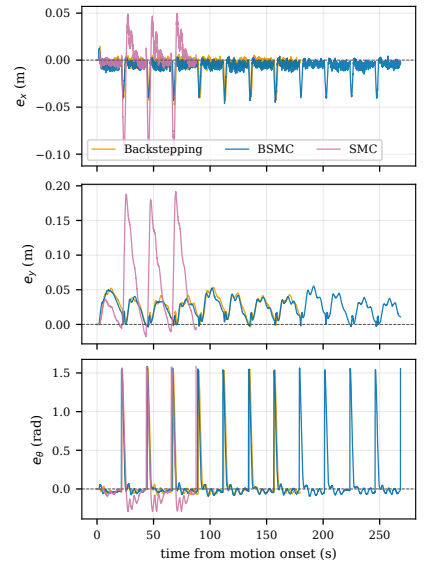
Fig. 2. Measured physical trajectory tracking for Backstepping, BSMC, and SMC. Each trajectory is expressed in its rigidly aligned local frame. The square SMC trace is a tuning pilot rather than a final validation run.



(a) Circle error histories.

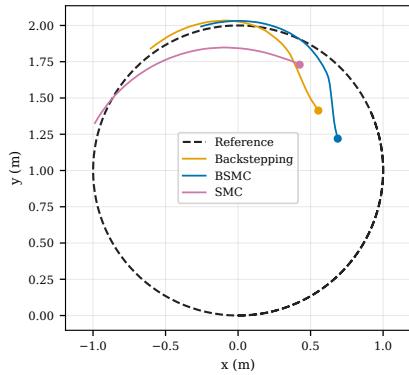


(b) Figure-eight error histories.

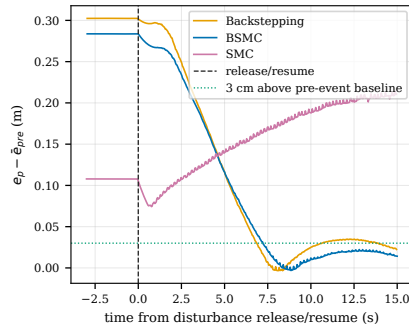


(c) Square error histories.

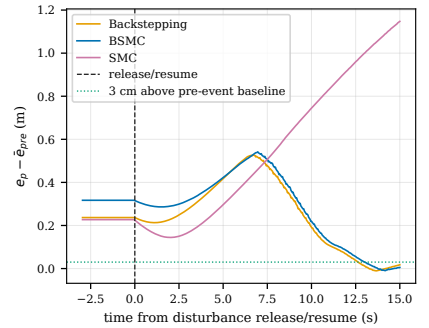
Fig. 3. Measured position and heading error histories for the selected nominal physical runs. The periodic square heading peaks coincide with the commanded 90° corner transitions.



(a) Yaw- 0° disturbed trajectories.



(b) Yaw- 0° recovery histories.



(c) Yaw- 90° recovery histories.

Fig. 4. Physical manual-disturbance pilots on the circular path. A missing recovery value denotes failure to satisfy the fixed recovery criterion within the recorded interval. The selected yaw- 90° SMC trial was safety-censored.

V. CONCLUSION AND FUTURE WORK

The selected physical runs confirm closed-loop operation of the complete STM32F411–ESP-NOW–ROS 2–AprilTag system. BSMC and Backstepping provide similar performance on the selected circle and square runs; BSMC gives the lowest selected square position RMSE, while Backstepping gives the lowest circle RMSE and straight-square heading RMSE. The selected pure-SMC square pilot is limited by large post-corner transients. Manual displacement tests show that robust global relocalization and reliable inner wheel-speed tracking are prerequisites for meaningful controller comparison.

The principal limitation is the unbalanced nature of the current pilot data. Future work will freeze controller parameters, restore and verify the inner wheel-speed loop, and collect equal-repeat validation runs under identical firmware, localization, initial-pose, battery, and floor conditions. Only that locked dataset should be used for mean–standard-deviation claims.

REFERENCES

- [1] Y. Kanayama, Y. Kimura, F. Miyazaki, and T. Noguchi, “A stable tracking control method for an autonomous mobile robot,” in *Proc. IEEE ICRA*, 1990, pp. 384–389.
- [2] V. Utkin, J. Guldner, and J. Shi, *Sliding Mode Control in Electromechanical Systems*, 2nd ed., CRC Press, 2009.
- [3] E. Olson, “AprilTag: A robust and flexible visual fiducial system,” in *Proc. IEEE ICRA*, 2011, pp. 3400–3407.
- [4] S. Macenski, T. Foote, B. Gerkey, C. Lalancette, and W. Woodall, “Robot Operating System 2: Design, architecture, and uses in the wild,” *Science Robotics*, vol. 7, no. 66, 2022.